

AGREEGATE FUNCTIONS(WEEK2)

Name:K.Vaishali

Roll Number: 2520030117

Step 1: Create Database

The screenshot shows the MySQL Workbench interface for a local instance named 'K.Vaishali Local instanceAG...'. The left sidebar contains navigation options under 'MANAGEMENT' (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), 'INCE' (Startup / Shutdown, Server Logs, Options File), and 'PERFORMANCE' (Dashboard, Performance Reports, Performance Schema Setup). Below these are 'Administration' and 'Schemas' tabs. The main editor window, titled 'MySQL13', contains two SQL statements: 1. CREATE DATABASE IF NOT EXISTS bank_transactions; and 2. USE bank_transactions;. The bottom panel shows the 'Output' tab with 'Action Output' selected, displaying a table of execution results.

#	Time	Action	Message
✓ 1	00:35:46	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected
✓ 2	00:35:46	USE bank_transactions	0 row(s) affected

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Step 2: Create Table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tab with a tree view of the database structure. The main editor window shows a SQL script for creating a database and a table. The script is as follows:

```
1 CREATE DATABASE IF NOT EXISTS bank_transactions;
2 USE bank_transactions;
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
```

The right sidebar shows the 'SQL Snippets' panel. The bottom panel shows the 'Output' window with the following results:

#	Time	Action	Message
1	00:35:46	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected
2	00:35:46	USE bank_transactions	0 row(s) affected
3	00:36:44	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected
4	00:36:44	USE bank_transactions	0 row(s) affected
5	00:36:44	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected

Step 3: Insert Data Question 1: Sum()

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tab with a tree view of the database structure. The main editor window shows a SQL script for inserting data into the bank_transactions table. The script is as follows:

```
16 branch_name VARCHAR(50),
17 transaction_type VARCHAR(20),
18 amount DECIMAL(10,2),
19 transaction_date DATE
20 );
21
22 INSERT INTO bank_transactions VALUES
23 (101,'vaishali goud','Hyderabad','Deposit',5000,'2024-01-05'),
24 (102,'vaishali goud','Hyderabad','Withdrawal',2000,'2024-01-06'),
25 (103,'el','Vijayawada','Deposit',12000,'2024-01-08'),
26 (104,'sike','Vizag','Deposit',8000,'2024-01-10'),
27 (105,'dustin','Hyderabad','Withdrawal',3500,'2024-01-11'),
28 (106,'sadie sink','Vizag','Deposit',15000,'2024-01-12'),
29 (107,'kruthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
30 (108,'humsini','Hyderabad','Deposit',9000,'2024-01-14'),
31 (109,'bhavana','Vizag','Withdrawal',4000,'2024-01-15'),
32 (110,'tejaswini','Vijayawada','Deposit',11000,'2024-01-16');
33
34 USE bank_transactions;
35 SELECT * FROM bank_transactions;
```

The right sidebar shows the 'SQL Snippets' panel. The bottom panel shows the 'Output' window with the following results:

#	Time	Action	Message
9	00:51:34	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050. Table 'bank_transactions' a
10	00:51:38	INSERT INTO bank_transactions VALUES (101,'vaishali goud','Hyderabad','Deposit',5000,'2024-01-05'), (102...	10 row(s) affected Records: 10 Duplicates: 0
11	00:52:01	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
12	00:52:30	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected, 1 warning(s): 1007 Can't cre
13	00:52:30	USE bank_transactions	0 row(s) affected
14	00:52:30	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050. Table 'bank_transactions' a

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Question 2: Avg()

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
33 • USE bank_transactions;
34 • SELECT * FROM bank_transactions;
35 • SELECT SUM(amount) AS Total_Amount
36 • FROM bank_transactions;
37
38 • SELECT AVG(amount) AS Average_Transaction
39 • FROM bank_transactions;
```

The Results tab is selected, showing a single row with the value 7050.000000 for the column Average_Transaction.

Question 3: Max()

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
41 • FROM bank_transactions;
42
43 • SELECT AVG(amount) AS Average_Transaction
44 • FROM bank_transactions;
45
46 • SELECT MAX(amount) AS Highest_Transaction
47 • FROM bank_transactions;
```

The Results tab is selected, showing a single row with the value 15000.00 for the column Highest_Transaction.

Below the Results tab, the Output tab is selected, showing a log of actions and messages:

#	Time	Action	Message
14	00:52:30	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050, Tab
15	00:53:01	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected, 1 w
16	00:53:01	USE bank_transactions	0 row(s) affected
17	00:53:01	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050, Tab
18	00:53:03	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
19	00:53:49	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions	1 row(s) returned

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Question 4: Min()

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
52 FROM bank_transactions;
53
54 • SELECT MAX(amount) AS Highest_Transaction
55 FROM bank_transactions;
56
57 • SELECT MIN(amount) AS Lowest_Transaction
58 FROM bank_transactions;
```

The Results tab shows the output of the third query:

Lowest_Transaction
1000.00

The Output tab shows the execution log:

#	Time	Action	Message
15	00:53:01	CREATE DATABASE IF NOT EXISTS bank_transactions	1 row(s) affected
16	00:53:01	USE bank_transactions	0 row(s) affected
17	00:53:01	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050
18	00:53:03	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
19	00:53:49	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions	1 row(s) returned
20	00:54:30	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions	1 row(s) returned

Question 5: Count()

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
66 FROM bank_transactions;
67
68 • SELECT MIN(amount) AS Lowest_Transaction
69 FROM bank_transactions;
70
71 • SELECT COUNT(*) AS Total_Transactions
72 FROM bank_transactions;
```

The Results tab shows the output of the third query:

Total_Transactions
10

The Output tab shows the execution log:

#	Time	Action	Message
16	00:53:01	USE bank_transactions	0 row(s) affected
17	00:53:01	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050
18	00:53:03	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
19	00:53:49	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions	1 row(s) returned
20	00:54:30	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions	1 row(s) returned
21	00:55:16	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions	1 row(s) returned

AGREEGATE FUNCTIONS(WEEK2)

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Question 6: Where + Sum()

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
72 FROM bank_transactions;  
73 • SELECT COUNT(*) AS Total_Transactions  
74 FROM bank_transactions;  
75  
76 • SELECT SUM(amount) AS Total_Deposit  
77 FROM bank_transactions  
78 WHERE transaction_type = 'Deposit';
```

The Results tab shows the output of the query:

Total_Deposit
60000.00

The Output tab shows the execution log:

#	Time	Action	Message
17	00:53:01	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	Error Code: 1050. 1
18	00:53:03	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
19	00:53:49	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions	1 row(s) returned
20	00:54:30	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions	1 row(s) returned
21	00:55:16	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions	1 row(s) returned
22	00:56:15	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'	1 row(s) returned

Question 7: Group By

The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```
76 • SELECT SUM(amount) AS Total_Deposit  
77 FROM bank_transactions  
78 WHERE transaction_type = 'Deposit';  
79 • SELECT branch_name,  
80 SUM(amount) AS Total_Amount  
81 FROM bank_transactions  
82 GROUP BY branch_name;
```

The Results tab shows the output of the query:

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

The Output tab shows the execution log:

#	Time	Action	Message
18	00:53:03	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions	1 row(s) returned
19	00:53:49	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions	1 row(s) returned
20	00:54:30	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions	1 row(s) returned
21	00:55:16	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions	1 row(s) returned
22	00:56:15	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'	1 row(s) returned
23	00:57:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned

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Question 8: Group By + Having

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
81 FROM bank_transactions
82 GROUP BY branch_name;
83 SELECT branch_name,
84 SUM(amount) AS Total_Amount
85 FROM bank_transactions
86 GROUP BY branch_name
87 HAVING SUM(amount) > 20000;
```

The Result Grid shows the output of the query:

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

The Output pane shows the execution log with the following entries:

- 19 00:53:49 SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions 1 row(s) returned
- 20 00:54:30 SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions 1 row(s) returned
- 21 00:55:16 SELECT COUNT(*) AS Total_Transactions FROM bank_transactions 1 row(s) returned
- 22 00:56:15 SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' 1 row(s) returned
- 23 00:57:18 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name 3 row(s) returned
- 24 00:58:32 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name 2 row(s) returned

Question 9: Group By + Order By

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
86 GROUP BY branch_name
87 HAVING SUM(amount) > 20000;
88 SELECT branch_name,
89 SUM(amount) AS Total_Amount
90 FROM bank_transactions
91 GROUP BY branch_name
92 ORDER BY Total_Amount DESC;
```

The Result Grid shows the output of the query:

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

The Output pane shows the execution log with the following entries:

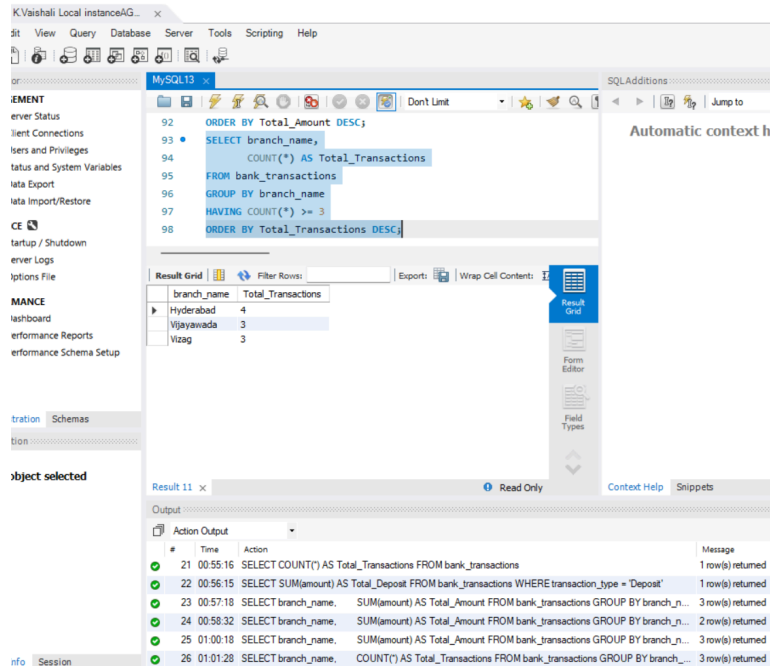
- 20 00:54:30 SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions 1 row(s) returned
- 21 00:55:16 SELECT COUNT(*) AS Total_Transactions FROM bank_transactions 1 row(s) returned
- 22 00:56:15 SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit' 1 row(s) returned
- 23 00:57:18 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name 3 row(s) returned
- 24 00:58:32 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name 2 row(s) returned
- 25 01:00:18 SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name 3 row(s) returned

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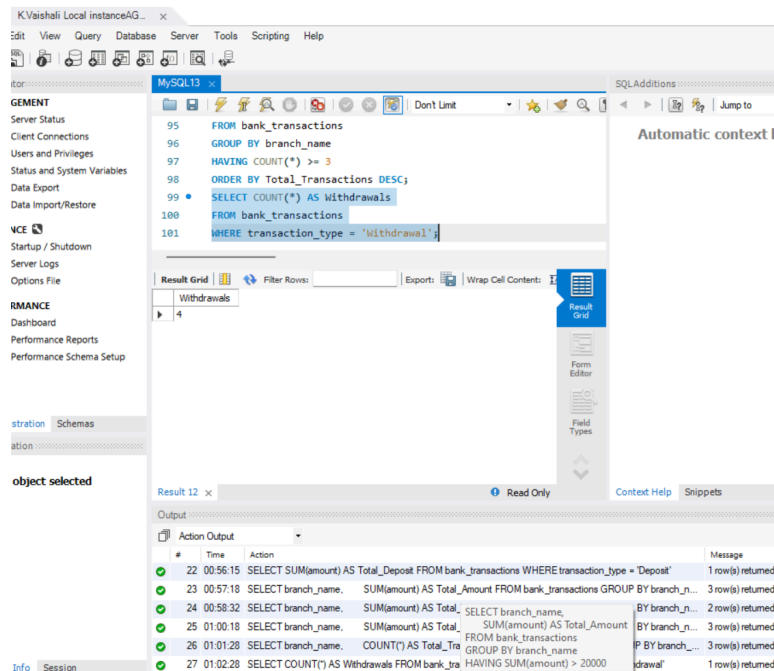
Question 10: Group By + Having + Order By



```
92 ORDER BY Total_Amount DESC;
93 SELECT branch_name,
94        COUNT(*) AS Total_Transactions
95 FROM bank_transactions
96 GROUP BY branch_name
97 HAVING COUNT(*) >= 3
98 ORDER BY Total_Transactions DESC;
```

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

Question 11: Where + Count()



```
95 FROM bank_transactions
96 GROUP BY branch_name
97 HAVING COUNT(*) >= 3
98 ORDER BY Total_Transactions DESC;
99 SELECT COUNT(*) AS Withdrawals
100 FROM bank_transactions
101 WHERE transaction_type = 'Withdrawal';
```

Withdrawals
4

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Question 12: Where + Avg()

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
98 ORDER BY Total_Transactions DESC;
99 • SELECT COUNT(*) AS Withdrawals
100 FROM bank_transactions
101 WHERE transaction_type = 'Withdrawal';
102 • SELECT AVG(amount) AS Avg_Deposit
103 FROM bank_transactions
104 WHERE transaction_type = 'Deposit';
```

The 'Result Grid' shows the result of the query:

Avg_Deposit
10000.000000

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message
23	00:57:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
24	00:58:32	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
25	01:00:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
26	01:01:28	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
27	01:02:28	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal'	1 row(s) returned
28	01:03:42	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'	1 row(s) returned

Question 13: Group By + Max()

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
102 • SELECT AVG(amount) AS Avg_Deposit
103 FROM bank_transactions
104 WHERE transaction_type = 'Deposit';
105 • SELECT branch_name,
106 MAX(amount) AS Highest_Amount
107 FROM bank_transactions
108 GROUP BY branch_name;
```

The 'Result Grid' shows the result of the query:

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

The 'Output' pane shows the execution log with the following entries:

#	Time	Action	Message
24	00:58:32	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	2 row(s) returned
25	01:00:18	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_n...	3 row(s) returned
26	01:01:28	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_...	3 row(s) returned
27	01:02:28	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type = 'Withdrawal'	1 row(s) returned
28	01:03:42	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type = 'Deposit'	1 row(s) returned
29	01:04:38	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_...	3 row(s) returned

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Question 14: Group By + Min()

```
106 MAX(amount) AS Highest_Amount
107 FROM bank_transactions
108 GROUP BY branch_name;
109 SELECT branch_name,
110 MIN(amount) AS Lowest_Amount
111 FROM bank_transactions
112 GROUP BY branch_name;
```

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Question 15: Where + Order By

```
110 MIN(amount) AS Lowest_Amount
111 FROM bank_transactions
112 GROUP BY branch_name;
113 SELECT *
114 FROM bank_transactions
115 WHERE amount > 8000
116 ORDER BY amount DESC;
```

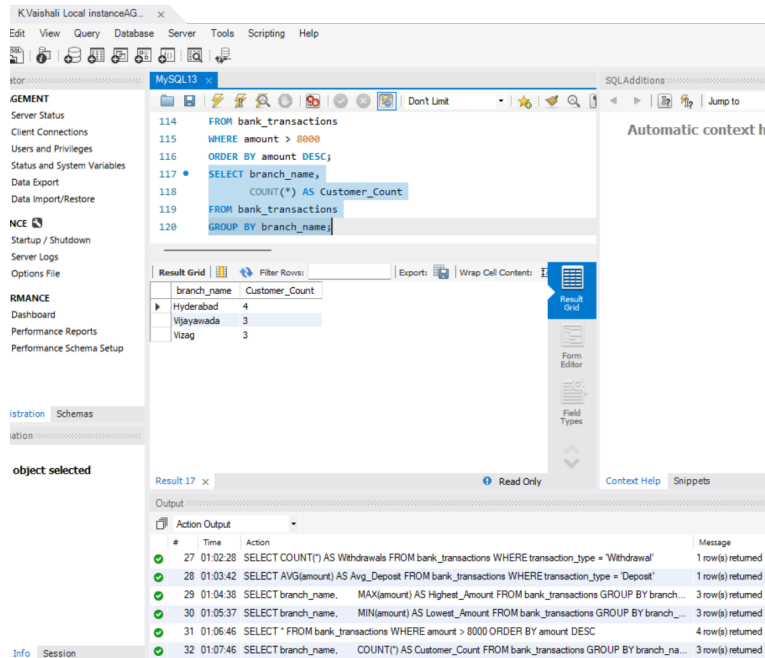
txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	sadie sink	Vizag	Deposit	15000.00	2024-01-12
103	el	Vijayawada	Deposit	12000.00	2024-01-08
110	tejaswini	Vijayawada	Deposit	11000.00	2024-01-16
108	humini	Hyderabad	Deposit	9000.00	2024-01-14

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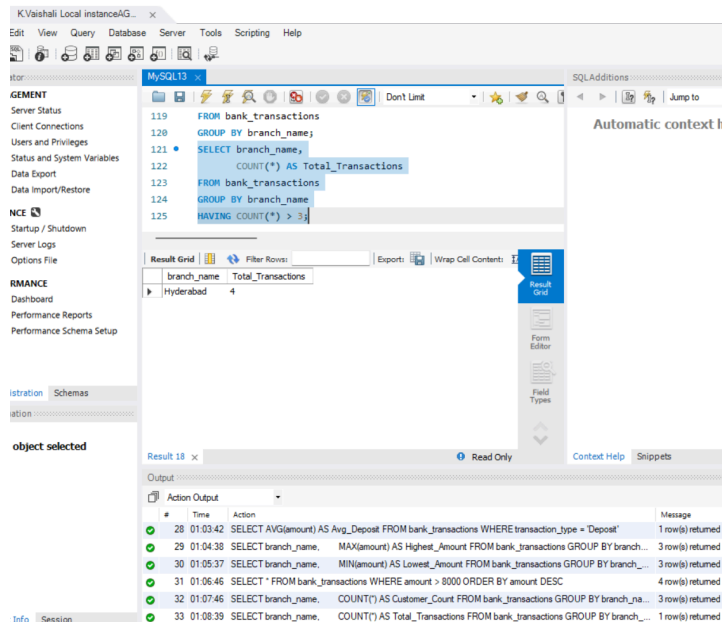
Question 16: Group By + Count()



```
114 FROM bank_transactions
115 WHERE amount > 8000
116 ORDER BY amount DESC;
117 SELECT branch_name,
118        COUNT(*) AS Customer_Count
119 FROM bank_transactions
120 GROUP BY branch_name;
```

branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

Question 17: Having + Count()



```
119 FROM bank_transactions
120 GROUP BY branch_name;
121 SELECT branch_name,
122        COUNT(*) AS Total_Transactions
123 FROM bank_transactions
124 GROUP BY branch_name
125 HAVING COUNT(*) > 3;
```

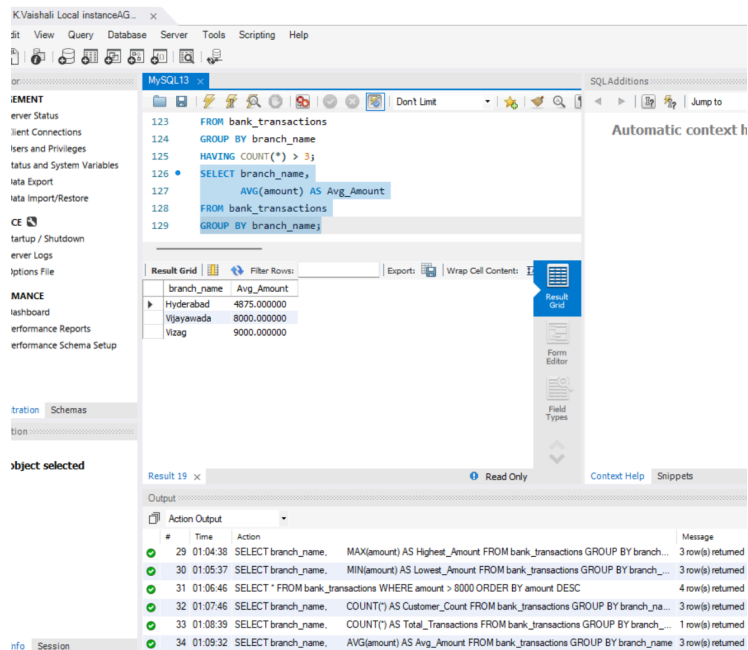
branch_name	Total_Transactions
Hyderabad	4

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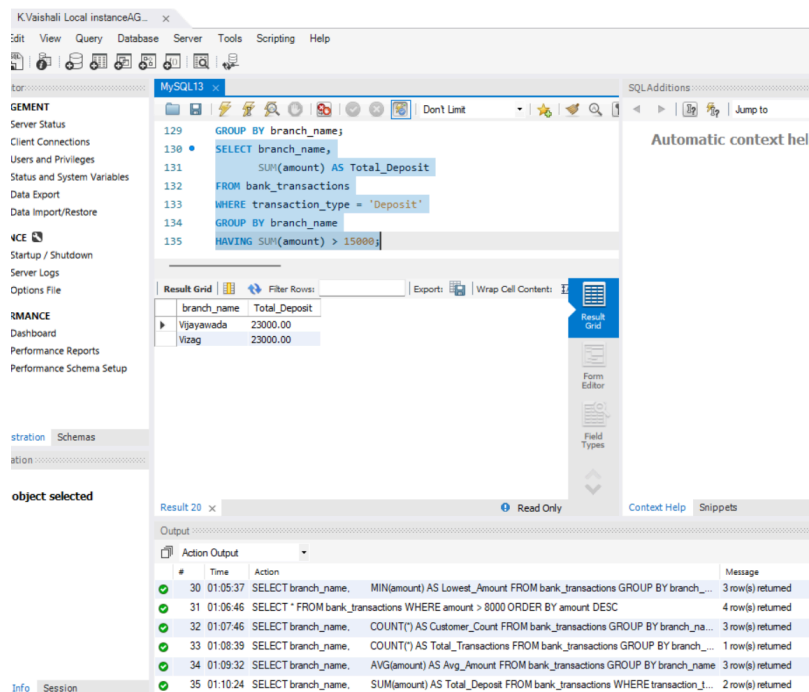
Question 18: Group By + Avg()



```
123 FROM bank_transactions
124 GROUP BY branch_name
125 HAVING COUNT(*) > 3;
126 SELECT branch_name,
127        AVG(amount) AS Avg_Amount
128 FROM bank_transactions
129 GROUP BY branch_name;
```

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

Question 19: Where + Group By + Having



```
129 GROUP BY branch_name;
130 SELECT branch_name,
131        SUM(amount) AS Total_Deposit
132 FROM bank_transactions
133 WHERE transaction_type = 'Deposit'
134 GROUP BY branch_name
135 HAVING SUM(amount) > 15000;
```

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

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Question 20: Group By + Having + Order By

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
135 HAVING SUM(amount) > 15000;  
136 SELECT branch_name,  
137        AVG(amount) AS Avg_Amount  
138 FROM bank_transactions  
139 GROUP BY branch_name  
140 HAVING AVG(amount) > 7000  
141 ORDER BY Avg_Amount DESC;
```

The Results tab shows the following data:

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

The Output tab shows the Action Output log:

#	Time	Action	Message
31	01:06:46	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC	4 row(s) returned
32	01:07:46	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name	3 row(s) returned
33	01:08:39	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name	1 row(s) returned
34	01:09:32	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name	3 row(s) returned
35	01:10:24	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_id > 1000	2 row(s) returned
36	01:11:17	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name	2 row(s) returned